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Peer-graded Assignment: Week 6: SVM Lab

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Week 6: SVM Lab

by Anonymous Learner
October 23, 2023

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PROMPT

In **Homework 6 part 1B**, you were asked:

Which training examples are the **support vectors**? Assign the coordinate in the list. e.g. support_vectors = [[1,0],[0,0]]

Explain your answer for which training examples are support vectors. For the support vectors that you have identified, which are the **in-bound support vectors** and which are the **out-bound support vectors**?

Type your answer in the rich text field.

The support vectors for the SVM model that I've identify were [6, 3], [5, 2], [6, 0], and [5, 1] . [6, 3] and [5, 2] are in-bound support vectors, correctly classified and inside the margin. [6, 0] and [5, 1] are out-bound support vectors, lying on the margin boundary but not contributing to misclassification and [6, 3], [5, 2] are in-bound support vectors, correctly classified and inside the margin.

RUBRIC

Here is a copy of the visualization from the notebook (to help the solution make sense):

Here is a sample solution:

From the data it is a fair assumption that red points above the decision boundary are correctly classified and blue points below the decision boundary are correctly classified. From this we see that (-5,0) and (5,2) are the in-bound support vectors and (-5,1) and (6,3) are the out-bound support vectors.

- 3 pts
  Properly explains answer for which training examples are support vectors **and** identifies (-5,0) and (5,2) as in-bound support vectors **and** identifies (-5,1) and (6,3) as out-bound support vectors.
- 2 pts
  Two of the following correct, one incorrect: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.
- 1 pt
  One of the following correct, two incorrect: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.
- 0 pts
  Doesn't correctly identify any of the following: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.

PROMPT

Homework 6 Part 1C asks the following question:

How did you know which training examples had nonzero slack?

Type your answer in the rich text field.

Outbound support vectors, such as (-5,1) and (6,3), are on the wrong side of their support vector boundaries, resulting in nonzero slack. These are the points that contribute to the margin violations.

RUBRIC

Here is the solution:

The training examples that are the outbound support vectors, (-5,1) and (6,3) are on the wrong side of their support vector boundaries,
so they are the only ones with nonzero slack.

- 2 pts
  Identifies that the outbound support vectors (-5, 1) and (6,3) are the only ones with nonzero slack because they are on the wrong side of their support vector boundaries.
- 1 pt
  Identifies that the support vectors with nonzero slack are on the wrong side of their support vector boundaries but doesn't identify that the outbound support vectors (-5, 1) and (6,3) are the only ones with nonzero slack.
- 0 pts
  Doesn't identify that the outbound support vectors (-5,1) and (6,3) are the only ones with nonzero slack or they are on the wrong side of their support vector boundaries.

PROMPT

In **Homework 6, Part 1B** has the following directions:

Experiment with **different values of C**. Answer the following question in your Peer Review for the week: How does the choice of C affect the nature of the decision boundary and the associated margin?

Type your answer in the rich text field.

The margin becomes wider when the C parameter is smaller. Smaller C values allow for more slack in the model, leading to a wider soft margin.

RUBRIC

The solution is:

The margin becomes larger when C is smaller.

- 5 pts
  Properly identifies how the choice of C affects the nature of the decision boundary **and** that the margin becomes larger when C is smaller.
- 3 pts
  Missing one of the following: properly identifies how the choice of C affects the nature of the decision boundary **or** that the margin becomes larger when C is smaller.
- 0 pts
  Missing both of the following: properly identifies how the choice of C affects the nature of the decision boundary **and** that the margin becomes larger when C is smaller.

PROMPT

Homework 6 Part 2C has the following directions:

Set C=3 and compare the results you get when using the hinge vs the squared_hinge values for the loss parameter. In this week's Peer Review Assignment: Explain your observations. Compare hinge loss vs. squared hinge loss.

Type your answer in the rich text field.

Hinge loss penalizes support vectors linearly based on their distances from the hyperplane. In contrast, squared hinge loss more heavily penalizes support vectors that have larger violations. As a result, squared hinge loss provides more flexibility within the dashed lines but enforces fewer support vectors on the wrong side of the decision boundary compared to plain hinge loss.

RUBRIC

Here is a **sample solution:**

Hinge loss penalizes support vectors linearly to their distances to the hyperplane, whereas **squared hinge loss** more heavily penalizes ones that have more violations. Thus, squared hinge loss provides more slack within the dashed lines but allows fewer support vectors on the plane's wrong side than plain hinge loss.

- 5 pts
  Answer gives a thorough and correct comparison of hinge loss and squared hinge loss similar to the sample solution.
- 3 pts
  Answer gives a somewhat correct comparison of hinge loss and squared hinge loss- answer includes some of the same components as the sample solution. However, answer also includes some incorrect statements.
- 0 pts
  No answer or answer is completely incorrect.

PROMPT

Homework 6 Part 2D has the following directions:

In general, how does the choice of C affect the bias and variance of the model? Write your answer in this week's Peer Review assignment.

Type your answer in the rich text field.

A smaller C allows for more slack in the model, making it more robust to misclassifications and resulting in a simpler decision boundary.

RUBRIC

Here is a **sample solution**:

A small c can allow more slack and is therefore a robust model.

- 5 pts
  Answer gives thorough description of how the choice of C affects the bias and variance of the model.
- 3 pts
  Answer is partially correct but includes at least one incorrect statement.
- 0 pts
  Answer is completely incorrect.

PROMPT

Homework 6 part 3B has the following directions:

In this part we'll use cross-validation to estimate the validation accuracy achieved by our model. Experiment with the values of the hyperparameters to see if you can get a good validation accuracy. Describe your experiment strategy for the C hyperparamter. How does the choice of C affect the resulting decision boundary?

Type your answer in the rich text field.

Smaller C values create a wider soft margin, allowing more flexibility in the model and accommodating some misclassifications. In contrast, larger C values lead to a tighter soft margin and a more strict classification.

RUBRIC

Here is a C tuning code solution:

```
nlsvm = SVC(kernel='rbf', C=0.1, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel='rbf', C=1, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel='rbf', C=10, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
```

A smaller C allows more slack (wider soft margin). A larger C has tighter soft margin.

- 5 pts
  Includes all of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.
- 3 pts
  Missing at least one of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.
- 0 pts
  Doesn't include any of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.

PROMPT

Homework 6 part 3B has the following directions:

In this part we'll use cross-validation to estimate the validation accuracy achieved by our model. Experiment with the values of the hyperparameters to see if you can get a good validation accuracy. Describe your experiment strategy for the gamma hyperparamter. How does the choice of gamma affect the resulting decision boundary?

Type your answer in the rich text field.

A larger gamma parameter results in a more complex decision boundary, potentially overfitting the data by making it too sensitive to individual data points.

RUBRIC

Here is a gamma tuning solution:

```
nlsvm = SVC(kernel='rbf', C=1, gamma=5)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel='rbf', C=1, gamma=10)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel='rbf', C=1, gamma=50)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
```

A larger gamma leads to having a more complex decision boundary

- 5 pts
  Includes all of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.
- 3 pts
  Missing at least one of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.
- 0 pts
  Doesn't include any of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.

PROMPT

Homework 6 Part 3C has the following directions:

How does the choice of kernel function affect the bias/variance of the model?

Type your answer in the rich text field.

The choice of kernel function in SVMs impacts bias and variance:

- Linear: Higher bias, lower variance.
- RBF: Lower bias, higher variance.
- Polynomial: Bias depends on degree; higher degrees lower bias.
- Sigmoid: Higher bias, moderate variance.

A flexible kernel, such as the RBF or a high-order polynomial, allows for a more complex model with increased variance but reduced bias.

RUBRIC

Here is a sample solution:

An expressible (flexible) kernel (such as RBF or polynomial with high order) makes a more complex model- increases variance, reduces bias.

- 5 pts
  Learner's solution is similar to sample. Properly and adequately describes how the choice of kernel function affects the bias/variance of the model.
- 3 pts
  Learner's solution is partially correct but includes at least one incorrect statement.
- 0 pts
  Learner's solution is completely incorrect.

PROMPT

Homework 6 Part 4A has the following coding directions:

Replace it with a proper GridSearchCV object and run the grid search with cross validation.

Please upload a screenshot of your code for this section.

GridSearchCV with cross validation code

```
# Use sklearn.svm.SVC() and GridSearchCV
from sklearn.svm import SVC
# Define the parameter grid to search over (hyperlinks) range
# Hint: Use logspace(-5, 5, 11, base=2) for C and logspace(-5, 5, 11, base=2) for gamma
param_grid = {'C': np.logspace(-5, 5, 11, base=2),
              'gamma': np.logspace(-5, 5, 11, base=2)}
# Create an SVM model with RBF kernel
svm = SVC(kernel='rbf')
# Create a GridSearchCV object with 5-fold cross-validation (cv=5)
grid_search = GridSearchCV(estimator=svm, param_grid=param_grid, cv=5)
# Fit the grid search to the data
grid_search.fit(X, y)
# Print the best hyperparameters and the corresponding accuracy
best_params = grid_search.best_params_
best_accuracy = grid_search.best_score_
print("Best C: {}, Best gamma: {}".format(best_params['C'],
print("Best cross-validation accuracy: {:.2f}".format(best_accuracy))
```

```
Best C: 5, Best gamma: 1
Best cross-validation accuracy: 0.943
```

RUBRIC

Here is a solution:

```
c_range = np.logspace(-5, 5, 11, base=2)
g_range = np.logspace(-5, 5, 11, base=2)
param_grid = dict(gamma=g_range, C=c_range)
grid = GridSearchCV(SVC(kernel='rbf'), param_grid=param_grid, cv=5)
grid.fit(X, y)
```

- 5 pts
  Includes both of the following: Learner creates a proper GridSearchCV object **and** runs the grid search with cross validation.
- 3 pts
  Learner missing one of the following: learner creates a proper GridSearchCV object **or** runs the grid search with cross validation.
- 0 pts
  Missing both of the following: Learner creates a proper GridSearchCV object **and** runs the grid search with cross validation.

PROMPT

Homework 6, Part 4B has the following heat-map question:

The following function will plot a heat-map of the cross-validation accuracies for each combination of parameters. Which combination looks the best? How did you use the heat-map to select the best combination?

Type your answer in the rich text field.

By visualizing the heat-map, look for the combination with the whitest (highest) color, which represents the best cross-validation accuracy. In the context of the heat-map, the combination of parameters where the color is closest to white or yellow (indicating a high mean test score) is typically considered the best. This corresponds to the optimal C and gamma values that result in the highest accuracy on the cross-validation data. The combinations (C=8, gamma=8), (C=16, gamma=8), and (C=32, gamma=2) achieve the highest validation accuracy. These parameter combinations seem to perform best in terms of accuracy.

RUBRIC

Here is a solution:

By looking at the grid visualization, (C=8, gamma=8), (C=16,gamma=8) and (C=32, gamma=2) have highest validation accuracy.

- 5 pts
  Includes both of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) and learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).
- 3 pts
  Includes one of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) **or** learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).
- 0 pts
  Doesn't include either of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) **or** learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).

PROMPT

Homework 6 Part 4C says:

The GridSearchCV object scores, among other things, the best combination of parameters as well as the cross-validation accuracy achieved with those parameters. Print those quantities for our model.

Submit a screenshot of best parameters and best accuracy here.

best parameters and best accuracy

```
# Use sklearn.svm.SVC() and GridSearchCV
from sklearn.svm import SVC
# Define the parameter grid to search over (hyperlinks) range
param_grid = {'C': np.logspace(-5, 5, 11, base=2),
              'gamma': np.logspace(-5, 5, 11, base=2)}
# Create an SVM model with RBF kernel
svm = SVC(kernel='rbf')
# Create a GridSearchCV object with 5-fold cross-validation (cv=5)
grid_search = GridSearchCV(estimator=svm, param_grid=param_grid, cv=5)
# Fit the grid search to the data
grid_search.fit(X, y)
# Print the best hyperparameters and the corresponding accuracy
best_params = grid_search.best_params_
best_accuracy = grid_search.best_score_
print("Best C: {}, Best gamma: {}".format(best_params['C'],
print("Best cross-validation accuracy: {:.2f}".format(best_accuracy))
```

```
Best C: 5, Best gamma: 1
Best cross-validation accuracy: 0.943
```

RUBRIC

Here is a solution:

```
beat parameters:  {'C': 8.0, 'gamma': 8.0}
beat accuracy:  0.9433333333333334
```

Did the learner submit a similar screenshot of best parameters and best accuracy?

- 5 pts
  Yes
- 0 pts
  No

PROMPT

Part 4D has the following directions:

The GridSearchCV object also stores the classifier trained with the best hyperparameters. Pass this **best estimator** into the **nonlinear_plot** function to view the best decision boundary.

Upload a screenshot of your code.

best decision boundary code

```
In [38]: # pass GridSearchCV object best estimator into nonlinear_plot
         # Hint: Use grid.best_estimator_
         # grid_search: GridSearchCV object
         # X_train, y_train: training data
         # Assuming the best decision boundary using the nonlinear_plot function
         nonlinear_plot(X_train, y_train, grid_search.best_estimator_)
```

RUBRIC

Here is a sample solution:

```
nonlinear_plot(X, y, grid.best_estimator_)
```

Evaluate the learner's submission.

- 2 pts
  Learner calls nonlinear plot function **and** passes in X, y, and the best estimator stored in the GridSearchCV object similar to the solution.
- 1 pt
  Learner calls nonlinear plot function. Does not properly pass in X, y, and the best estimator stored in the GridSearchCV object similar to the solution
- 0 pts
  Learner does not call nonlinear plot function.

PROMPT

Upload a screenshot or copy of your plot for 4D.

The lab directions for 4D are as follows:

The GridSearchCV object also stores the classifier trained with the best hyperparameters. Pass this **best estimator** into the **nonlinear_plot** function to view the best decision boundary.

In the prompt above, you uploaded a screenshot of your code. Now **upload a screenshot or copy of your plot.**

best decision boundary plot

RUBRIC

Here is a sample solution:

- 2 pts
  Learner submitted a nonlinear plot showing the decision boundary from passing the best_estimator into the nonlinear_plot function.
- 0 pts
  Learner did not submit a nonlinear plot showing the decision boundary from passing the best_estimator into the nonlinear_plot function.

PROMPT

For your final question for 4D:

Based on the plot you generated in 4D (and submitted above), what can you tell about the best decision boundary?

Type your answer in the rich text field.

It appears that the decision boundary is highly flexible, adapting to the data's distribution with many degrees of freedom. However, it seems that the margin (the space between the decision boundary and the data points) is relatively wide, implying that only a few data points are influential in defining the model. This suggests that the majority of the data points are not significantly affecting the decision boundary's position, possibly due to a limited number of support vectors, which are the data points closest to the decision boundary and critical for the model's performance.

RUBRIC

Evaluate the learner's response about the best decision boundary

- 1 pt
  Learner's response makes sense based on the plot that they submitted.
- 0 pts
  Learner's response does not make sense based on the plot that they submitted.

Submit Review

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